# 4: SOFTWARE

[CHAPTER 4]

MST_CREATOR

Book Link

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# Note To Reader

These notes are made according to the 2023 – 2025 computer science syllabus so if you have your examinations for 2022, you may want to refer to other places.

Another thing is that I would advise you to use other resources to revise just to be safe (these notes may be missing information here and there).

I would also advise you to practice past papers as they are helpful when preparing!

Anyway, if this helps you, I am happy for you 😊

That's it for this note section thing so you can go to the actual content.

# Software Types & Interrupts

## [1] System & Application Software

Software can be split into system & application software:

### System Software

*Examples:*

1. Utilities
2. Operating System (OS)
3. Device Drivers

*Functions:*

1. Set of programs to manage & control operations of computer hardware.
2. Provides a platform on which other software can run.
3. Allow hardware & software to run without problems.
4. Provides human computer interface (HCI).
5. Controls allocation & usage of hardware resources.

### Application Software

*Examples:*

1. Photo editors
2. Video editors
3. Word processors

*Functions:*

1. Used to perform many tasks (apps) on a computer.
2. Allows user to perform certain tasks using the computer's resources.
3. May be a single program (like Notepad).
4. User can execute it when needed

# Utility Software (Utilities)

Many utility programs are on the computer system, some can be run by the user and some run in the background:

1. Virus Checkers
2. Defragmentation Software
3. Disk content analysis & repair
4. File compression & management
5. Back-up software
6. Security
7. Screen savers

## Virus Checkers (Anti-Virus Software)

These are programs that scan and protect a computer from malware and virus attacks in the background.

Common features of anti-virus software include:

1. Check software or files before they are run or loaded.
2. Compares a possible virus against a database with known viruses.
3. Checks software to check for suspicious activity that would be done by a virus (heuristic checking).
4. Files that are suspicious are put into quarantine (meaning that the virus will be automatically deleted).
    - Also allows user to decide whether to keep or delete the file as user may know if the file is not a virus (this is a false positive).
5. Full systems are carried out at least once a week.
6. Need to be updated every now and then as new viruses keep getting discovered.

# Defragmentation Software

This software takes all the data fragments scattered around on the hard drive and rearranges them into contiguous sectors.

This decreases read/write time and HDD head movements.

# Backup Software

What it does:

1. Allow a schedule for backing up files to be made.
2. Only carry out backup when a file has been changed.

There are 3 version of a file for complete security:

1. Current version: Stored in internal SSD or HDD.
2. Locally backed up copy: Stored in other places (like portable SSD).
3. Remote backed up copy: Stored away from machine (like on cloud).

| OS | Windows | Mac OS |
| --- | --- | --- |
| **Utility Name** | File History | Time Machine |
| **What It Does** | Takes snapshots of files and stores them on external HDD regularly, this builds a library of past file versions which allows the user to choose which version they want to back up. | Erases contents of selected drive and replaces them with back up. It is necessary to connect external drive is connected. |
| **How Often It Saves** | Every hour (default) | Every hour<br>Daily back-up for past month<br>Weekly back-ups for previous months |

# Security Software

What it does:

1. Manages user accounts and control.
2. Links into other utility software.
3. Protects network interfaces.
4. Uses encryption & decryption to ensure that intercepted data is useless without the decryption key.
5. Oversees updating of software.

# Screen Savers

They are programs that blank the screen or fill it with moving images when the computer has been idle for a certain amount of time.

They are a way of customization and a way to maintain security as the computer can be locked after a certain amount of time.

Screen savers also allow other programs to run in the background such as:

1. Virus scans
2. Distributed computer apps

# Device Drivers

They are software that allow the communication between OS and peripheral devices (like mouse or keyboard).

# [2] Operating Systems (OS)

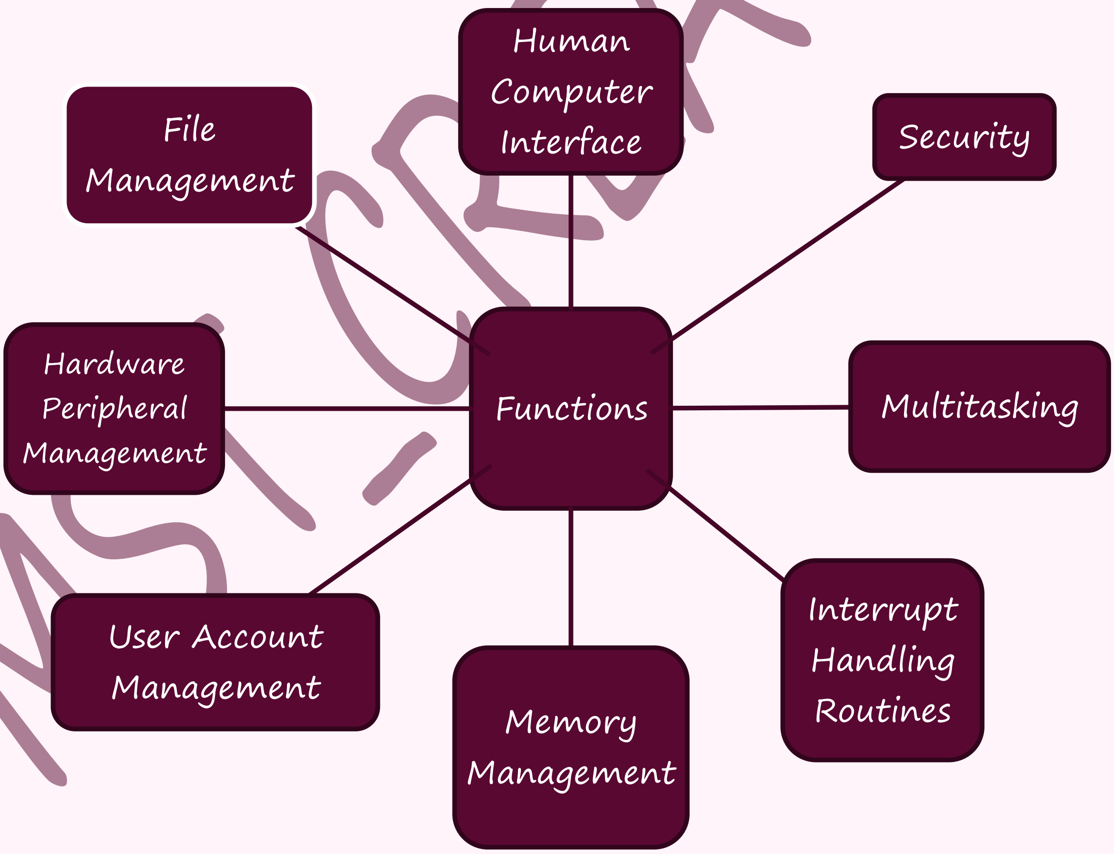
Operating systems are software that allow computers to function correctly and allow users to communicate with the system.

OS provides an environment for applications to run and provide an interface for users to easily use computers.

Computers store the OS on the HDD or the SSD as they are very large programs.

Operating systems of mobile phones are stored on their SSD (they don't have HDD).

Functions of an OS:



# Human Computer Interface (HCI)

The HCI is in the form of Command Line Interface (CLI) & Graphical User Interface (GUI).

| **Name** | **CLI** | **GUI** |
|---|---|---|
| **What It Is** | This requires the user to type in commands to tell the computer what to do. | This requires user to click on logos and images to tell the computer what to do. |
| **Advantages** | 1. User communicates directly with the computer.<br>2. User is not restricted to predetermined options.<br>3. Possible to alter computer configuration settings.<br>4. Uses small amount of computer memory. | 1. User does not need to learn any commands.<br>2. More user friendly.<br>3. Simple as we use a pointing device to click on an icon to carry out a task. |
| **Disadvantages** | 1. User needs to learn many commands to carry out basic tasks.<br>2. Commands must be typed; this is prone to errors.<br>3. Each command must be correctly spelt, formatted, etc. | 1. Uses more memory than CLI.<br>2. User is limited to icons present on the screen.<br>3. Needs an OS which consumes a lot of storage. |

# Memory Management

Functions:

1. Manages primary storage (RAM) and allows movement of data between RAM and HDD/SSD when executing a program.
2. Keeps track of all memory locations.
3. Carries out memory protection so applications do not store data in the same memory slot as this could cause data loss, incorrect results (from application), etc.

# Security Management

Ensures the integrity, confidentiality, and availability of data by:

1. Carrying out OS updates when available.
2. Ensuring anti-virus and other security software is up to date.
3. Maintaining access rights for all users.
4. Making use of privileges to prevent users from entering certain areas on a computer that permits multi-user activity.
5. Helping prevent illegal intrusion into your system.
6. Offering ability to recover data when it has been lost or corrupted.



# Hardware Peripheral Management

1. Communicates with all input & output devices using device drivers.
2. Uses device drivers to take data from a file which is defined by the OS and translates it, so the input/output device understands it.
3. Ensures that each hardware resource has a priority so they can be used and released when needed.
4. Manages input & output devices by controlling queues & buffers.

# File Management

Main Tasks:

1. File naming conventions which can be used (filename.docx).
2. Perform certain tasks on a file (like delete or copy).
3. Maintains the directory structures.
4. Ensures access control mechanisms are maintained (like password protection).
5. Ensures memory allocation for a file by reading it from HDD/SSD and loading it into the memory.

# Multitasking

This allows computers to carry out more than one task at a time.

Each process taking place uses a fraction of the hardware resource, this is monitored by the OS to ensure that the processes are running properly:

1. Resources are allocated to a process for a specific amount of time.
2. The process can be interrupted while running.
3. Processes are organized into priority levels so they can use resources according to the priority.

Multitasking is also helpful as the main memory, HDD/SSD and virtual memory are better managed, this is an effective use of CPU time.

# User Accounts Management

Computers allow multiple users to use one computer, this means that the data of each user must be kept separate from the other.

When this happens, each user is allowed to:

1. Customise their screen layouts and other settings.
2. Use separate files and folders and manage them by themselves.

## Admins

The management of the user accounts is generally overseed by an admin.

The admin/administrator can create and delete users and can also restrict their activity on the system.

# [3] Running of Applications

The OS provides application software a platform to run on.

## How It Works

When a computer starts:

1. Part of the OS is loaded into the RAM (this is called booting up or bootstrap loading).
2. The start-up of the computer's motherboard is handled by the BIOS (Basic Input/Output System).
3. The BIOS then tells the system where the OS can be found.
4. The BIOS then loads the part of the OS that is required to be executed.

# Firmware

The BIOS is referred to as firmware.

Firmware is a program that provides low level control for devices.

# EEPROM

The BIOS is stored in a special ROM called EEPROM which stands for "Electronically Erasable Programmable ROM".

The EEPROM is a flash memory chip which means that its contents remain the same even when the computer is turned off.

The only problem is that the user can rewrite, update, or even remove the BIOS.

**Note:** You do not need to know a lot about EEPROM, just this much is enough.



# CMOS

CMOS stands for "Complementary Metal Oxide Semi-conductor".

The CMOS chip stores the settings of the BIOS.

The CMOS is charged with the battery on the motherboard, this means that if the user had changed the settings, they will be set to the factory settings.

# In The End...

The application software will be under the control of the OS and will need to access system software when running. Different parts of the OS will need to be loaded from the RAM as the software runs.

# [4] Interrupts

An interrupt is a signal sent from a device or from a software to the microprocessor.

Interrupts allow computers to multitask.

The interrupt causes the microprocessor to stop temporarily and service the interrupt.

## What Causes Them?

Interrupts can be caused by:

1. **Timing signals**
2. **Input/output processes:** like printer requiring more data
3. **User interactions:** Like key presses or mouse movement
4. **Software errors that cause problems:** Like trying to divide by 0 or trying to access memory space that is being used by other software
5. **Hardware fault:** Like paper is jammed in the printer

## What Happens When Interrupt Is Received

Once the interrupt signal is received, the microprocessor:

1. Continue doing what it was doing
2. Stops and services the device or software that sent the signal.

To decide what to do, the computer establishes "interrupt priorities".

# Servicing The Interrupt

1. The computer first saves the task being run, this means that it saves the contents of the Program Counter (PC) and other registers.
2. The Interrupt Service Routine (ISR) is executed by loading the start address into the program counter (PC).
3. Once the interrupt is fully serviced, the status of the interrupted task is reinstated, all the register contents are received, and the task proceeds.

# Buffers

A buffer is a memory area that stores data temporarily.

# Programming Language Types & IDEs

## [1] High & Low-Level Languages

| Name | High Level Languages (HLL) | Low Level Languages (LLPL) |
|---|---|---|
| **Definition** | A language like the human language which programmers can easily use and understand. | A language that relates to specific architecture and hardware of a particular system from machine code, binary and hexadecimal. |
| **Examples** | Python, Java, C++, etc. | Machine code & Assembly Languages |
| **Advantages** | 1. Easier to read, write, debug, and understand the code.<br>2. Quicker to write programs.<br>3. Easier to maintain programs in use.<br>4. Programs can be run on any computer. | 1. Can make use of special hardware.<br>2. Incudes special, machine dependant instructions.<br>3. Programs are fast and don't take up much space. |
| **Disadvantages** | 1. Programs can be large.<br>2. Programs may take long to execute.<br>3. Programs may not be able to make use of special hardware. | 1. Takes long time to debug and write the programs.<br>2. Code is hard to understand. |

# [2] Assembly Languages

Assembly language is made up of mnemonics (like LDA, STO, etc).

| Mnemonic | Meaning |
|---|---|
| **LDA** | Gets the value of a variable and loads it into the accumulator. |
| **STO** | Replaces the value of a variable with the value stored in the accumulator. |
| **ADD** | Adds the value of a variable with the variable in the accumulator. |

# [3] Translators

A translator is a utility program that translates the program into binary so it can be run by a computer.

## Compilers, Interpreters, and Assemblers

| Name | Compiler | Assembler | Interpreter |
|---|---|---|---|
| **What it does** | Translate HLL program to machine code. | Translates assembly language program to machine code. | Executes a HLL program one line at a time. |
| **Action When Error is Found** | Generates an error report. | Generates an error report. | Stops running and outputs error message |
| **Executable Produced?** | Yes | Yes | No |
| **How Many Machine Code Instructions?** | One statement can be many instructions. | One statement is one instruction. | One statement may require execution of many instructions. |
| **Compiler Needed?** | Yes | Yes | No |
| **When Is It Used?** | Distributed for general use. | Distributed for general use. | When program is in development. |

# [4] Advantages & Disadvantages of Compilers & Interpreters

| Name | Compilers | Interpreters |
|---|---|---|
| **Advantages** | 1. Compiled programs can be stored ready to use.<br>2. Compiled programs can run without a compiler.<br>3. Compiled programs are executed in less time.<br>4. Compiled programs take less memory when executed. | 1. It is easy to edit the programs when in development.<br>2. Easier and quicker to debug & test programs in development. |
| **Disadvantages** | Takes long time to write, debug, and test programs during development. | 1. Programs cannot be run without interpreter.<br>2. Programs take longer to execute. |

# [5] IDEs

IDE stands for "Integrated Development Environment".

IDEs are used by programmers to aid the writing and development of programs.

Examples of IDEs include PyCharm, VS Code, Notepad++, etc.

## General Features

| Feature | Description |
|---|---|
| **Code Editor** | Allows program to be written and edited without needing separate text editor. |
| **Translator** | This can be an interpreter and/or a compiler to enable the program to be executed.<br>The interpreter is used for developing and the compiler is used to produce a final version which can be used. |
| **Runtime environment with debugger** | This used to check if there are any logic errors as this shows us how all the values of all the variables change in runtime. We can also use breakpoints which stope code execution at a certain point of the code. |
| **Error diagnostics & Auto correction** | Dynamic error checking finds possible errors in the code as we type, the error is pointed out and corrected automatically by the IDE. |
| **Auto Completion** | Context sensitive prompts with text completion for variable names, functions, and other reserved words. |
| **Auto-Documenter** | These are helpful guides that tell us about functions, syntax, etc. |
| **Pretty Typing** | This is when the IDE gives colour to the code in a meaningful way as it can show or categorise something (like functions are blue or strings are orange) |

# Thanks For Reading!

Share these if you think these will be helpful to anyone else!